JEE Mains 2020 Jan Chapter wise Question Bank

Area Under Curves

Q1

The area that is enclosed in the circle $x^2 + y^2 = 2$ which is not common area enclosed by $y = x$ & $y^2 = x$ is

वृत्त $x^2 + y^2 = 2$ के अन्तर्गत परिबद्ध वह क्षेत्रफल जो $y = x$ & $y^2 = x$ द्वारा परिबद्ध क्षेत्रफल के साथ उभयनिष्ठ न हो–

(1) $\frac{1}{12}(24\pi - 1)$ (2) $\frac{1}{6}(12\pi - 1)$ (3) $\frac{1}{12}(6\pi - 1)$ (4) $\frac{1}{12}(12\pi - 1)$

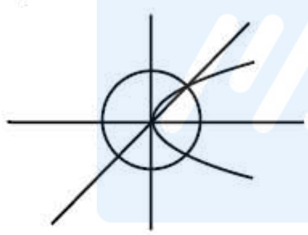
7th Jan Morning

Sol

(2)

Total area – enclosed area

कुल क्षेत्रफल – उभयनिष्ठ क्षेत्रफल



$$2\pi - \int_0^1 \sqrt{x} - x\,dx$$

$$2\pi - \left(\frac{2x^{3/2}}{3} - \frac{x^2}{2}\right)_0^1$$

$$2\pi - \left(\frac{2}{3} - \frac{1}{2}\right) \Rightarrow 2\pi - \left(\frac{1}{6}\right) \Rightarrow \frac{12\pi - 1}{6}$$

02

The area bounded by $4x^2 \le y \le 8x + 12$ is -

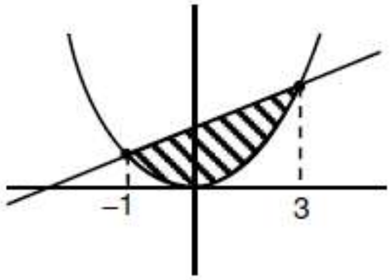
$4x^2 \le y \le 8x + 12$ द्वारा घिरा हुआ क्षेत्रफल है–

(1) $\frac{127}{3}$ (2) $\frac{128}{3}$ (3) $\frac{124}{3}$ (4) $\frac{125}{3}$

7th Jan Evening

Sol

(2)



$4x^2 = y$
$y = 8x + 12$
$4x^2 = 8x + 12$
$x^2 - x - 3 = 0$
$x^2 - 2x - 3 = 0]$
$x^2 - 3x + x - 3 = 0$
$(x + 1)(x - 3) = 0$
$x = -1$

$$A = \int_{-1}^{3} (8x + 12 - 4x^2)\, dx$$

$$A = \frac{8x^2}{2} + 12x - \frac{4x^3}{3}\Bigg|_{-1}^{3} = (4(9) + 36 - 36) - \left(4 - 12 + \frac{4}{3}\right) = 36 + 8 - \frac{4}{3}$$

$$= 44 - \frac{4}{3} = \frac{132 - 4}{3} = \frac{128}{3}$$

03

If $y^2 = ax$ and $x^2 = ay$ intersect at A & B. Area bounded by both curves is bisected by line $x = b$(given $a > b > 0$). Area of triangle formed by line AB, $x = b$ and x-axis is $\frac{1}{2}$. Then

(1) $a^6 - 12a^3 - 4 = 0$
(2) $a^6 + 12a^3 - 4 = 0$
(3) $a^6 - 12a^3 + 4 = 0$
(4) $a^6 + 12a^3 + 4 = 0$

8th Jan Morning

Sol

(3)

$$\int_0^b \left(\sqrt{a}x^{\frac{1}{2}} - \frac{x^2}{a} \right) dx = \frac{a^2}{6}$$

$\Rightarrow \quad \frac{2}{3}\sqrt{a}\, b^{\frac{3}{2}} - \frac{b^3}{3a} = \frac{a^2}{6} \quad$ ....(i)

also area of Δ OQR का क्षेत्रफल $= \frac{1}{2}$

$\frac{1}{2}b^2 = \frac{1}{2} \quad \Rightarrow \quad b = 1$

Put in (i) मे रखने पर

$\Rightarrow \quad 4a\sqrt{a} - 2 = a^3$

$\Rightarrow \quad a^6 + 4a^3 + 4 = 16a^3$

$\Rightarrow \quad a^6 - 12a^3 + 4 = 0$

04

Let P be the set of points (x, y) such that $x^2 \le y \le -2x + 3$. Then area of region bounded by points in set P is

माना P बिन्दुओं (x, y) का समुच्चय इस प्रकार है कि $x^2 \le y \le -2x + 3$. तो P के सभी बिन्दुओं द्वारा सम्बद्ध क्षेत्रफल होगा:

(1) $\frac{16}{3}$ (2) $\frac{32}{3}$ (3) $\frac{29}{3}$ (4) $\frac{20}{3}$

8th Jan Evening

Sol

(2)

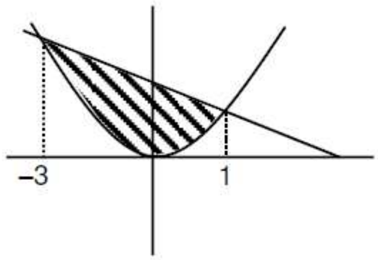
Point of intersection of $y = x^2$ & $y = -2x + 3$ is

$y = x^2$ तथा $y = -2x + 3$ का प्रतिच्छेदन बिन्दु

obtained by प्राप्त होता है, $x^2 + 2x - 3 = 0$

$\Rightarrow x = -3, 1$

So, Area अतः क्षेत्रफल $= \int_{-3}^{1}(3-2x-x^2)dx = 3(4) - 2\left(\frac{1^2-3^2}{2}\right) - \left(\frac{1^3+3^3}{3}\right) = 12 + 8 - \frac{28}{3} = \frac{32}{3}$



05

If $f(x) = \begin{cases} x & 0 < x < \frac{1}{2} \\ \frac{1}{2} & x = \frac{1}{2} \\ 1-x & \frac{1}{2} < x < 1 \end{cases}$

$g(x) = \left(x - \frac{1}{2}\right)^2$ then find the area bounded by f(x) and g(x) from $x = \frac{1}{2}$ to $x = \frac{\sqrt{3}}{2}$.

यदि $f(x) = \begin{cases} x & 0 < x < \frac{1}{2} \\ \frac{1}{2} & x = \frac{1}{2} \\ 1-x & \frac{1}{2} < x < 1 \end{cases}$

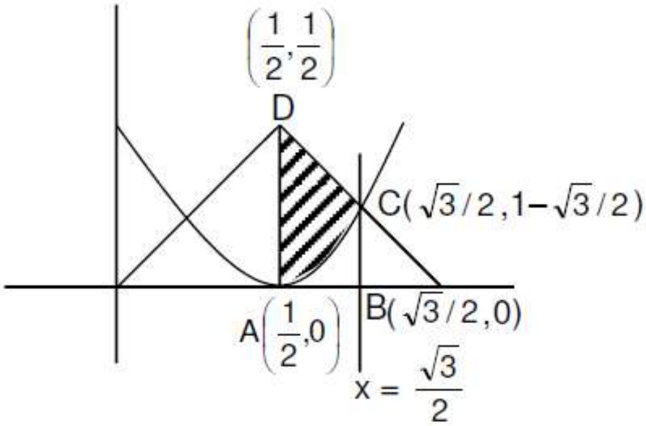
$g(x) = \left(x - \frac{1}{2}\right)^2$ तब f(x) तथा g(x) के द्वारा $x = \frac{1}{2}$ से $x = \frac{\sqrt{3}}{2}$ तक परिबद्ध क्षेत्रफल ज्ञात कीजिये। .

(1) $\frac{\sqrt{3}}{4} - \frac{1}{3}$ (2) $\frac{\sqrt{3}}{4} + \frac{1}{3}$ (3) $2\sqrt{3}$ (4) $3\sqrt{3}$

9th Jan Evening

Sol

(1)



Required area = Area of trapezium ABCD $- \int_{1/2}^{\sqrt{3}/2} \left(x - \frac{1}{2}\right)^2 dx$

$= \frac{1}{2}\left(\frac{\sqrt{3}-1}{2}\right)\left(\frac{1}{2}+1-\frac{\sqrt{3}}{2}\right) - \frac{1}{3}\left(\left(x-\frac{1}{2}\right)^3\right)_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$

$= \frac{\sqrt{3}}{4} - \frac{1}{3}$

mathongo

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Anup Sir